Note regarding the Inventory App

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At the moment, the “Inventory App” for the mobile phone, specifically the Android platform, is just in the “alpha” stage for development. The following are some notes regarding what was done, and what the mobile application is limited in,

- Regarding the login page and “new users,”
  - There are no role-based access controls or specific permissions given to a user, everyone with a user account can access the entire database shell, and anyone can make an account.
  - The passwords are not properly iterated or hashed, thus the accounts can be easily hacked if anyone has access to the SQLite database.
- Regarding the “database shell,”
  - An update function has not been implemented.
  - The database shell uses a `TableLayout` to render tabular data (and this layout still needs serious work done to display information properly).
  - A refresh button is added to refresh the table when new items are pushed into the database.
  - The table automatically refreshes after an item is removed, but not after an item is added (I do not know why this is the case).
- Regarding SMS notifications, I have developed a proof-of-concept, under the activity: `activity_sms.xml` and its respective Java class in `ActivitySMS.java`. It is just merely a proof-of-concept that is accessed through a “pseudo-harness” in the `MainActivity.java` class.
- Regarding the codebase, it was developed “haphazardly.” I sadly did not adhere to the best coding practices,
  - There is a bit of inconsistencies in how I wrote down names for variables, methods and classes.
  - Regarding AI usage, I did occasionally rely on Google’s “AI overview,” though I did try to cite when I cited Google AI overview, and the source material that it is deriving its solutions from.
  - The AI assistance explains in part the inconsistencies in coding style. While I did not directly copy and paste code snippets, I did rely on a “complete” solution and sadly did not do enough to fully integrate them to my codebase.
  - Further refactoring and better practices are needed to push this mobile app beyond the “proof-of-concept,” or at best “alpha” stage onto a more mature “beta” that can have its quality assessed by QA testers and select end-users (or the general public if possible and need be).

Much work is still needed before a launch can be put out there. In fact, some of the final project requirements could not be met, including,

- **App Code Design II.C.iii**: There is no mechanism to update items in the database
- **App Code Design III.B.i**: The SMS notifications exists purely as a proof-of-concept, I am not quite sure exactly how to make use of SMS notifications, as least not right now.
- **App Code Design IV.A.(i.-iii.)**: The coding conventions contain a degree of inconsistency, in part because I have not followed the “Don’t Repeat Yourself” (DRY) principles typically found in object oriented programming languages, but also in other programming paradigms.

Comments in the code hopefully reference enough tutorials and documentation that I relied on when developing the “alpha” stages of this inventory app.

Though I think that apart from these “quirks” and limitations, enough has been shown to meet enough of the requirements outlined in the project 3 guidelines.